

a) Execution Time & Throughput Analysis:

Table 1: Execution Time and Throughput Comparison Across 5 Request Endpoints (50 vs 100 Samples)

Target Web Endpoint	Samples	Min(ms)	Max(ms)	Avg(ms)	Throughput	Error %
Home Page (GET/)	50	19	112	39.24	0.51 req/s	0 %
	100	14	283	39.4	1.01 req/s	1 %
Notice Board (GET/notices)	50	115	484	287.74	0.51 req/s	68 %
	100	110	500	288.76	1.01 req/s	72 %
Course Catalogue (GET/courses)	50	187	384	276.28	0.51 req/s	90 %
	100	176	426	258.04	1.01 req/s	80 %
User Login (POST /login)	50	66	172	112.16	0.51 req/s	0 %
	100	62	258	111.42	1.01 req/s	2 %
Data API (GET/api/download/256)	50	28	343	44.5	0.51 req/s	2 %
	100	28	154	39.55	1.01 req/s	0 %

b) Bandwidth Analysis:

Table 2: Bandwidth Comparison Across 5 Request Endpoints (50 vs 100 Samples)

Target Web Endpoint	Samples	Received(KB/sec)	Sent(KB/sec)
Home Page (GET/)	50	1.21	0.18
	100	2.4	0.35
Notice Board (GET/notices)	50	3.73	0.2
	100	7.39	0.39
Course Catalogue (GET/courses)	50	22.06	0.2
	100	43.65	0.4
User Login (POST /login)	50	54.43	0.53
	100	107.77	1.06
Data API (GET/api/download/256)	50	130.83	0.26
	100	258.42	0.51
Total	50	264.63	1.88
	100	523.8	3.73

c) Assertion & Error Rate Analysis:

The Home Page, Login, and Data API handled the load easily. They always loaded in under 100ms, staying well below our limit, which gave them a perfect 0% error rate regardless of whether 50 or 100 users were active.

Both the Course (/courses) and the Notice Board (/notices) struggled with the load. Both endpoints consistently averaged between 258ms and 288ms, violating the 200ms threshold. As a result, the Notice Board failed roughly 70% of the time and the Courses failed between 80% and 90% of the time.

Because two of the five main endpoints are a bit slow for the 200ms requirement, the system experienced a high overall error rate of about 22% to 23%.

d) YouTube Video Link : [JMeter Assignment Walkthrough](#)

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